



Faculty of Engineering & Technology
Electrical & Computer Engineering Department

Circuits & Electronics Lab

ENEE2103

Experiment No.10 - The Operational Amplifier

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10.1 Abstract

The objective of this experiment is to explore the applications of operational amplifier circuits, including adders, voltage followers, comparators, integrators, and differentiators. And study these circuits properties, measure characteristics like voltage gain.

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10.2 Theory

10.2.1 Adding Application

An Adding amplifier circuit, which is a type of operational amplifier (op-amp) circuit that can combine multiple input voltages into a single output voltage. The circuit uses an operational amplifier (op-amp) which is a high-gain electronic voltage amplifier with a differential input and a single output. The op-amp has two input terminals, inverting and non-inverting. [1]

In the summing amplifier circuit, the input voltages are applied to the inverting terminal. And the output voltage becomes directly proportional to the sum of all the input voltages according to the equation: [1]

$$V_{out} = -[(R_F/R_{in})V_1 + (R_F/R_{in})V_2 + (R_F/R_{in})V_3]$$

Where:

- V_{out} is the output voltage
- R_F is the feedback resistor
- R_{in} is the input resistor

If each input voltage (V_1 , V_2 , $V_3...$) has its own resistor (R_{in}) of equal value the output voltage equation becomes:

$$V_{out} = -R_F/R_{in} * (V_1 + V_2 + V_3 + ...)$$

By changing the value of the resistors, you can modify the gain of each input voltage. If all the resistors are of the same value, the circuit produces an equal-weighted sum of the input voltages. [1]

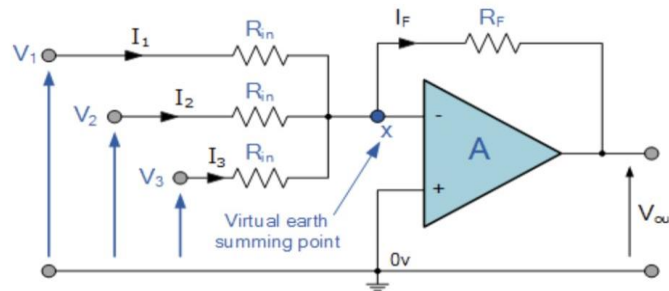


Figure 1: Adding circuit [1]

10.2.2 Voltage Follower Application

Voltage followers, also known as unity-gain amplifiers, buffer amplifiers, or isolation amplifiers. These circuits don't amplify the voltage signal. Instead, they act like a follower, keeping the output voltage exactly the same as the input voltage. Imagine feeding the op-amp 10 volts you'll get 10 volts right out. [2]

Voltage followers are essentially buffers. They provide a clean, undistorted copy of the input signal. They achieve this by having a very high input impedance, which means they don't draw much current from the source feeding the signal. This isolation protects the original signal from being weakened. [2]

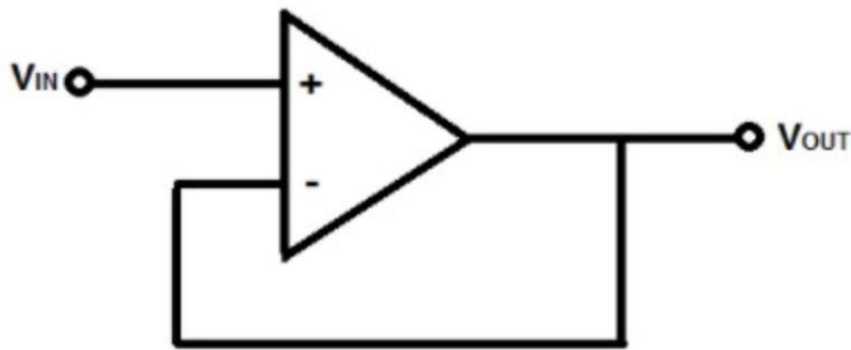


Figure 2: Voltage Follower circuit [2]

10.2.3 Comparator Application

The op-amp comparator acts like a voltage referee, deciding which input is higher. It can compare two analog voltages (V_{IN}) or an input voltage to a fixed reference voltage (V_{REF}). Unlike an amplifier, the comparator doesn't care about the exact voltage strength, only which one is bigger (V_{IN} or V_{REF}). and the circuit uses its high gain to rapidly switch its output voltage to either extreme of its power supply: positive ($+V_{CC}$) or negative ($-V_{CC}$). [3]

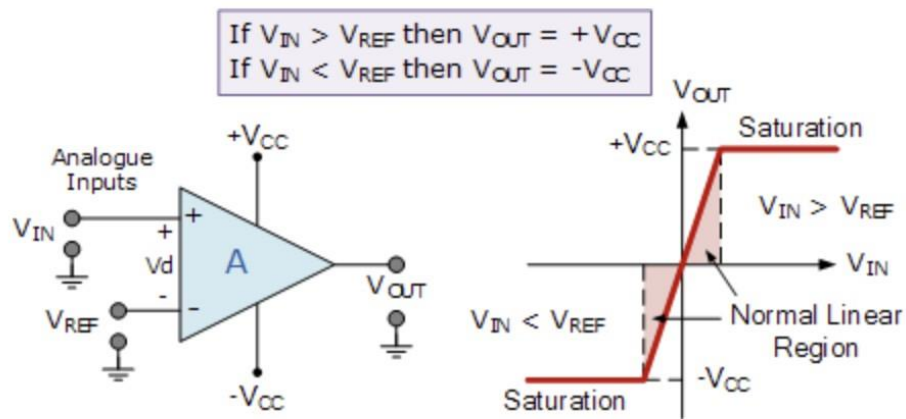


Figure 3: Comparator circuit [3]

- If V_{IN} is lower than V_{REF} . Since the positive input (V_{IN}) is lower than the negative input (V_{REF}), the output goes LOW, all the way down to the negative supply voltage ($-V_{CC}$).
- If you increase V_{IN} to be higher than V_{REF} , the output voltage quickly flips to HIGH, reaching the positive supply voltage ($+V_{CC}$). Even a slight decrease in V_{IN} below V_{REF} again triggers the output to switch back to LOW.

In essence, the comparator acts like a threshold detector. The output depends on V_{IN} 's level compared to V_{REF} : HIGH when V_{IN} is higher, LOW when V_{IN} is lower. This works regardless of which input (inverting or non-inverting) receives V_{IN} . [3]

10.2.4 Comparator with Hysteresis (Schmitt Trigger)

It adds a special feature to comparators. It creates two switching points instead of just one. Imagine a regular comparator that turns on (high output) when the input voltage reaches a certain point. With hysteresis, the comparator needs a higher voltage to turn on (upper threshold) but a lower voltage to turn off (lower threshold). This creates a zone between these two voltages where the output stays the same. [4]

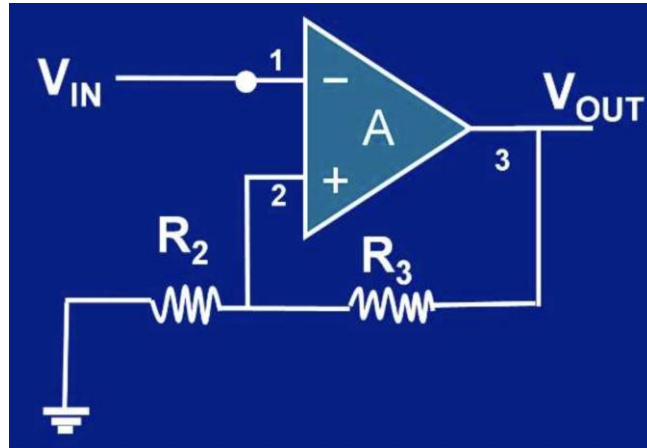


Figure 4: Hysteresis circuit [4]

The circuit uses an op-amp configured as an inverting comparator. The input voltage (V_{IN}) connects to the inverting input (V_-). Also, a voltage divider made of resistors R_2 and R_3 connects to the non-inverting input (V_+). This creates a reference voltage for comparison. A portion of the output voltage (V_{OUT}) is fed back to the positive feedback network (voltage divider). This creates a hysteresis. [4]

Example:

- Let's say the power supply is +5V and -5V, and both R_2 and R_3 have the same resistance value.
- Initially, the output (V_{OUT}) is at +5V (due to the high gain of the op-amp).
- As the input voltage (V_{IN}) increases, it needs to rise slightly above the reference voltage set by the divider (around +2.5V in this case) to trigger a switch.
- When V_{IN} goes above +2.5V, the output (V_{OUT}) flips to -5V.
- Now, the reference voltage for the divider changes because of the feedback. The new reference voltage is around -2.5V.
- As V_{IN} decreases, it needs to fall below -2.5V for the output to switch back to +5V.

This creates a "hysteresis zone" between +2.5V and -2.5V. Within this zone, the output voltage stays the same regardless of the input voltage changes. By adjusting the resistors R_2 and R_3 , you can control the width of this zone. [4]

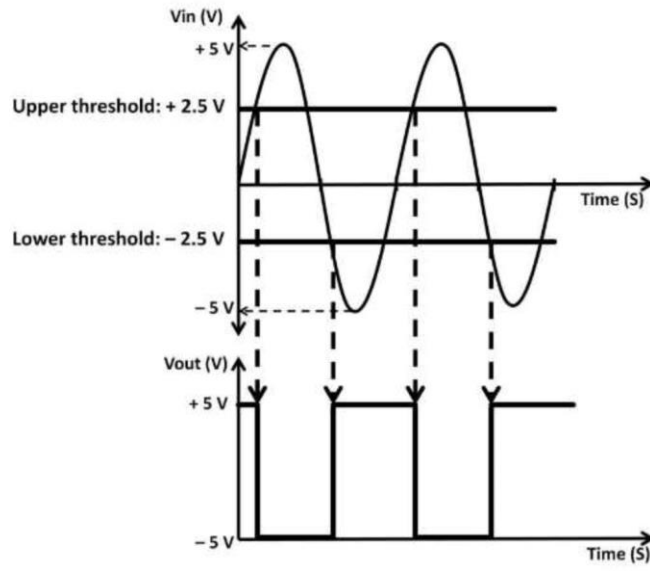


Figure 5: Hysteresis Output [4]

10.3 Procedure

10.3.1. Adding Application

The circuit of Figure (6) was set up with V1 controlled by the potentiometer and V2 obtained from the variable DC source on the trainer.

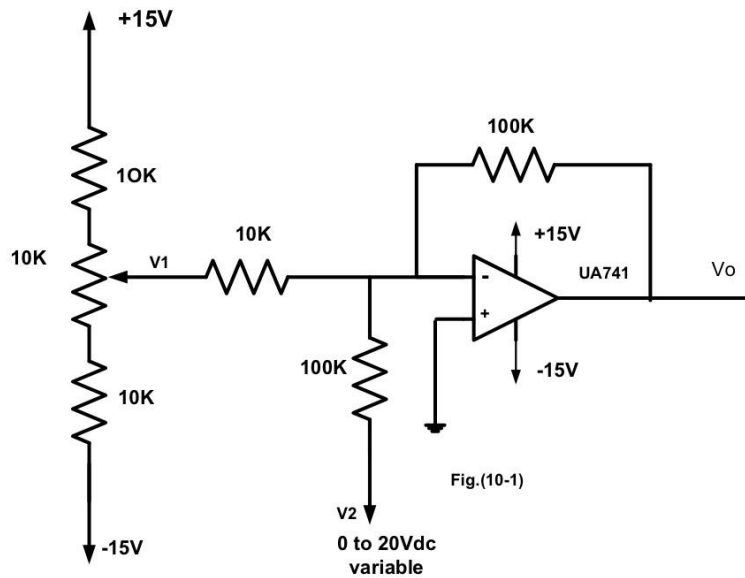


Figure 6: Adding circuit

The output voltage was measured for V1 and V2 as shown in Table 1:

Table 1: Adding circuit Output

Input voltage		Output Voltage	
V1	V2	V _o	Calculated voltage
0.498	1.988	-6.88	-6.968
0.301	4.018	-6.977	-7.028
-0.8983	2.006	6.858	6.977
-1.5197	6.008	8.955	9.189

and the expected output voltage for each step was calculated using the formula

$$V_{out} = -[(R_F/R_1)V_1 + (R_F/R_2)V_2]$$

$$(R_F/R_1) = (100k\Omega/10k\Omega) = 10$$

$$(R_F/R_2) = (100k\Omega/100k\Omega) = 1$$

$$V_{out} = -10V_1 - V_2$$

We observed that the theoretical values closely matched the experimental values. Due to the use of the inverting summing amplifier configuration in the circuit, when V_1 was negative, V_2 turned positive, and when V_1 was positive, V_2 turned negative.

10.3.2 Voltage Follower Application

The circuit of Figure (7) was set up.

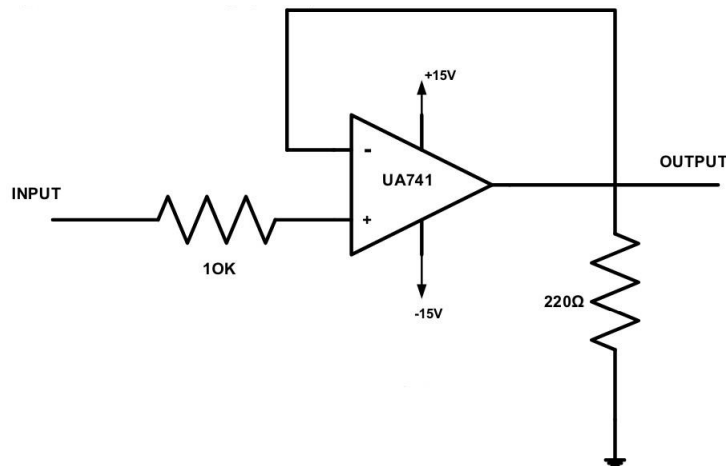


Figure 7: Voltage follower circuit

The output voltage (V_o) was measured and recorded for DC input values (V_i) of 1V, 3V, 4V, 5V, 6V, 7V, 8V, 10V, 12V, 14V, and 15V. The resistor R_L was then changed from 220Ω to $1K\Omega$, and the output voltage (V_o) was measured and recorded for the same input values. The results were documented in Table 2.

Table 2: Output voltage in different values for RL

Vi [DC,V]	1	3	4	5	6	7	8	10	12	14	15
Vo For RL = 220ohm	1	3	4	4.42	4.42	4.42	4.42	4.42	4.42	4.42	4.42
Vo For RL = 1kohm	1	3	4	5	6	7	8	10	12	12.21	12.22

Question:

- **Is this circuit has similar properties as the emitter follower. Explain ?**

Yes, this circuit has similar properties to an emitter follower. Both are used as voltage buffers, providing high input impedance and low output impedance. However, while the emitter follower provides slight voltage amplification, this circuit does not.

- **For what applications is this circuit used?**

Impedance matching, Signal isolation, Buffering ,Stabilizing voltage levels

- **What is the relation between your Vi, Vo?**

$V_o = V_i$ for $V_i < V_{sat}$

$V_o = -V_{sat}$ for $V_i > V_{sat}$

- **What is the approximate value of maximum output current of the op-amp?**

The approximate value of the maximum output current: $I_{out} = \frac{V_o}{R_L}$

$= \frac{4.42}{220} = 4.42 / 220 \Omega \approx 20.09\text{mA}$

10.3.3 Comparator Application

The circuit shown in Figure (8) was set up as instructed.

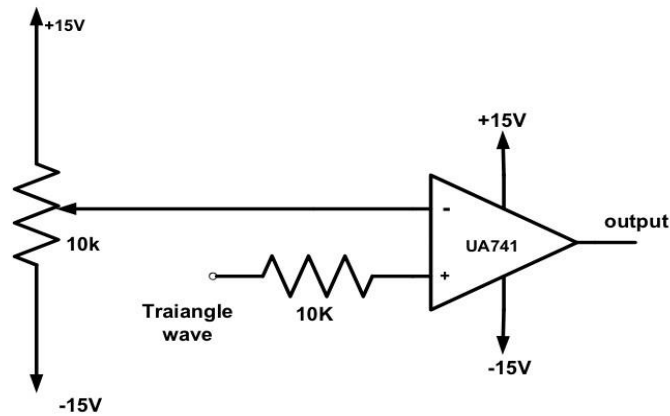


Figure 8: Comparator circuit

A 1 kHz triangular input signal was sourced from the function generator. The amplitude of the triangular input signal was set to 2 volts peak-to-peak. Then, the DC reference voltage was systematically adjusted in small increments. (This adjustment was aimed at achieving specific output states: positive saturation (+V_{sat}), negative saturation (-V_{sat}), and a square wave output). For each of these conditions, the output voltage was plotted, and the corresponding DC reference voltage was noted.

When $V_{dc} = -0.063 \text{ V}$



Figure 9: $V_{dc} = -0.063 \text{ V}$ in Comparator circuit

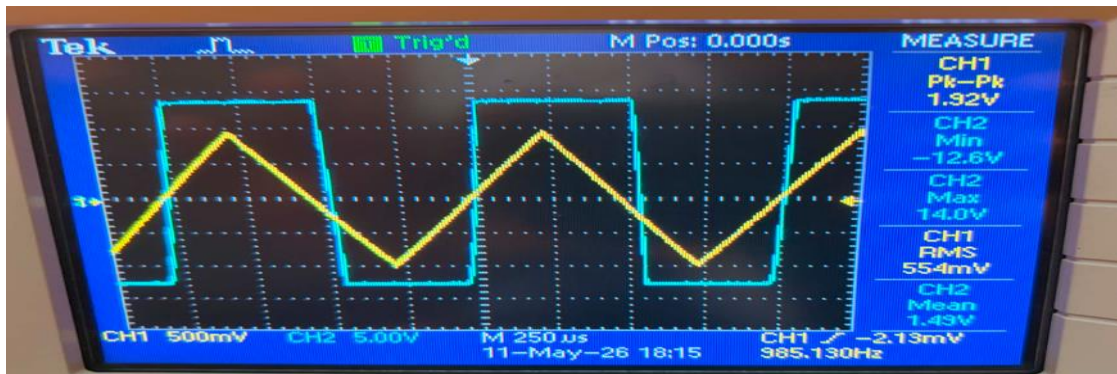


Figure 10: Comparator circuit output when $V_{dc} = -0.063 \text{ V}$

When the triangular input signal's amplitude (V_{ac}) is positive, the output of the circuit reaches positive saturation ($+V_{sat}$), Conversely, when V_{ac} is negative, the output reaches negative saturation ($-V_{sat}$),

When $V_{dc} = -1.479 \text{ V}$



Figure 11: $V_{dc} = -1.479 \text{ V}$ in Comparator circuit

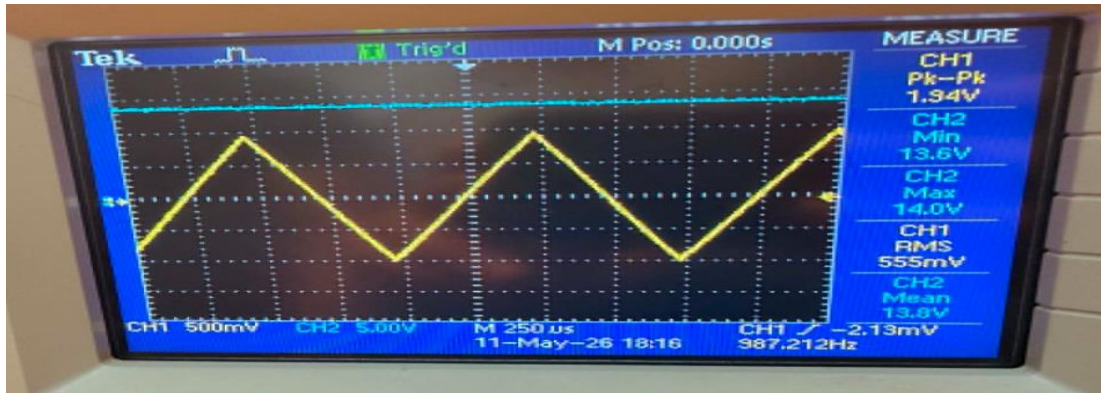


Figure 12: Comparator circuit output when $V_{dc} = -1.479 \text{ V}$

$+V_{sat} = 400 \text{ mV}$

When $V_{dc} = 1.538 \text{ V}$



Figure 13: $V_{dc} = 1.538 \text{ V}$ in Comparator circuit

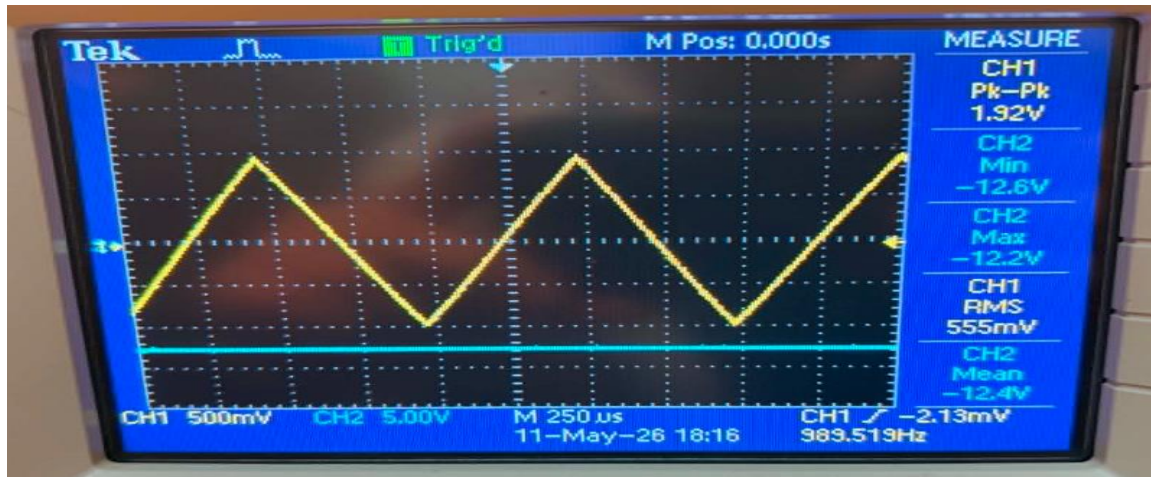


Figure 14: Comparator circuit output when $V_{dc} = 1.538V$

- $V_{sat} = 400mV$

Question:

• **What happens to the amplifier output?** the output voltage (V_o) is as follows:

- $V_o = +V_{sat}$ if $V_{ac} > V_{dc}$

- $V_o = -V_{sat}$ if $V_{ac} < V_{dc}$

- $V_o = 0$ if $V_{ac} = V_{dc}$

• **For what application is this circuit used?**

To check whether an input has reached some predetermined value.

• **Is there any similarity between this circuit and the diff amplifier, what is the shape of the output?**

While both the comparator and the differential amplifier use operational amplifiers, they differ in their output behavior. The comparator produces a square wave output, switching between positive and negative saturation based on input voltages' relationship. In contrast, the differential amplifier provides an output proportional to the voltage difference between its inputs.

10.3.4 Comparator with Hysteresis (Schmitt Trigger)

The Schmitt trigger circuit depicted in Figure (14) was connected as instructed.

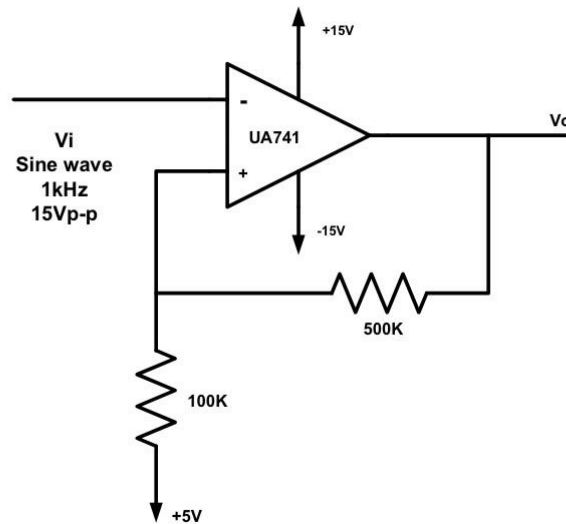


Figure 15: Hysteresis circuit

A 15V peak-to-peak sine wave with a frequency of 1 kHz was applied as the input signal (???)

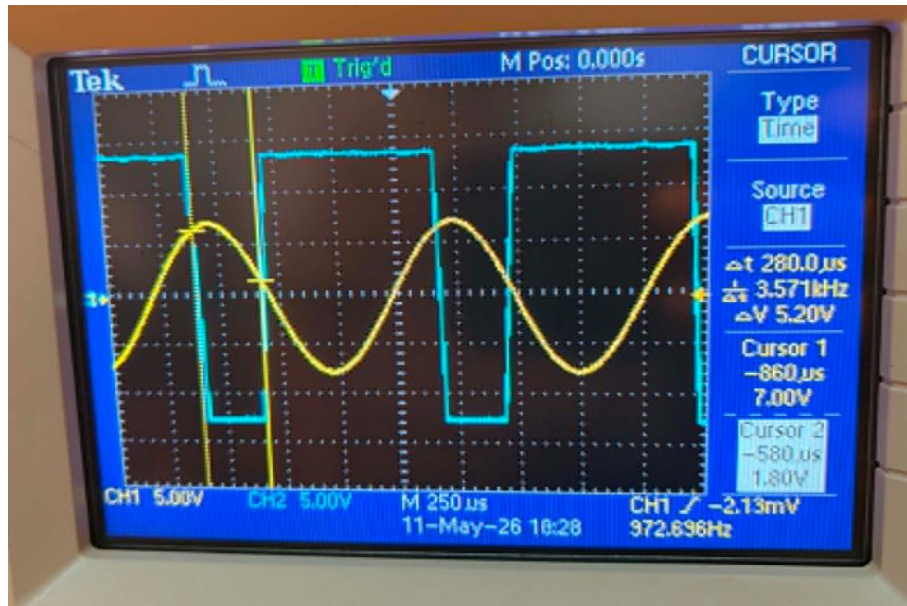


Figure 16: I/O voltage in Hysteresis circuit

As shown from Figure (16):

Upper threshold voltage (VUT) = 7.0 V

Lower threshold voltage (VLT) = 1.8 V

The theoretical lower and upper trigger levels for the circuit were calculated and compared with the measured values.

$V_o = +V_{sat} = 15 - 2 = 13V$, when $V_d > 0$ then $V(+)> V(-)$ where $V(-)= V_{in}$ So,
 $U_{th} = (100/100+500) * 13 + (500/ 100+500) * 5 = 6.32 V$

$V_o = -V_{sat} = -15 + 2 = -13V$, when $V_d < 0$ then $V(+)< V(-)$ where $V(-)= V_{in}$ So,
 $L_{th} = (100/100+500) * -13 + (500/100+500) * 5 = 2 V$ The

experimentally and theoretically values are close.

The oscilloscope was set to XY mode with the horizontal axis representing the input voltage and the vertical axis representing the output voltage. The resulting plot, shown in Figure (17), displays the hysteresis loop of the Schmitt Trigger.

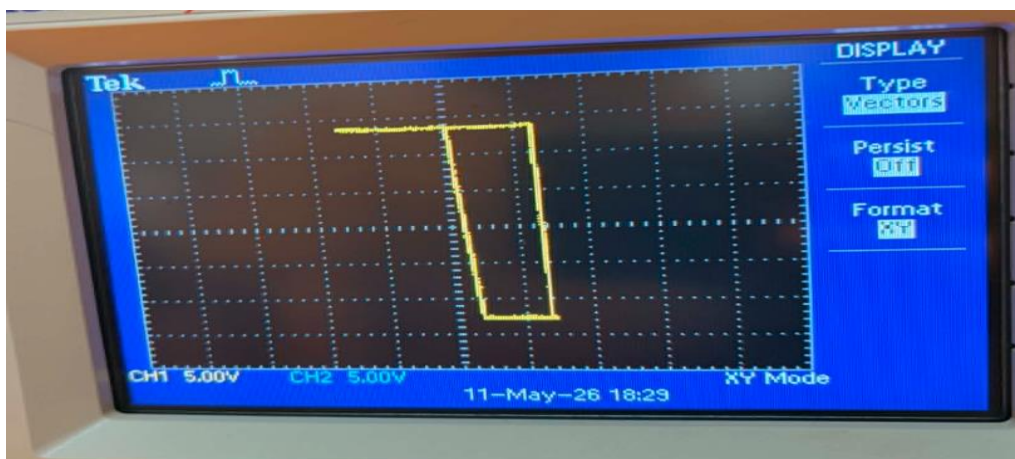


Figure 17: V_o as function of V_i using XY format.

10.4 conclusion

In this experiment, we explored various operational amplifier (op-amp) circuits, including adders, voltage followers, comparators, and Schmitt triggers. The adder circuit successfully combined multiple input voltages into a single output, with experimental results closely matching theoretical values. The voltage follower demonstrated high input impedance and low output impedance, effectively buffering input signals without amplification. The comparator differentiated input voltage levels, switching the output between positive and negative saturation, making it useful for threshold detection. The Schmitt trigger introduced hysteresis, creating stable digital signals from noisy inputs, with experimental thresholds matching theoretical predictions. Overall, the experiment showcased the versatility and practical applications of op-amp circuits in electronic systems.

10.5 References

- [1] https://www.electronics-tutorials.ws/opamp/opamp_4.html [Accessed 22/5/2026 12:13 AM]
- [2] <https://www.utmel.com/blog/categories/amplifiers/voltage-follower-purpose-working-and-circuits> [Accessed 22/5/2026 1:30 PM]
- [3] <https://www.electronics-tutorials.ws/opamp/op-amp-comparator.html> [Accessed 22/5/2026 2:15PM]
- [4] <https://developerhelp.microchip.com/xwiki/bin/view/products/amplifiers-linear/operational-amplifier-ics/introduction/comparators-with-hysteresis-schmitt-trigger/> [Accessed 22/5/2026 2:45PM]

10.6 Appendix

Experiment #10

ENEE2103

The Operational Amplifier

Objectives:

To investigate the application of the op. amp circuits such as adding, Voltage follower, Comparator, Integrator, and Differentiator.

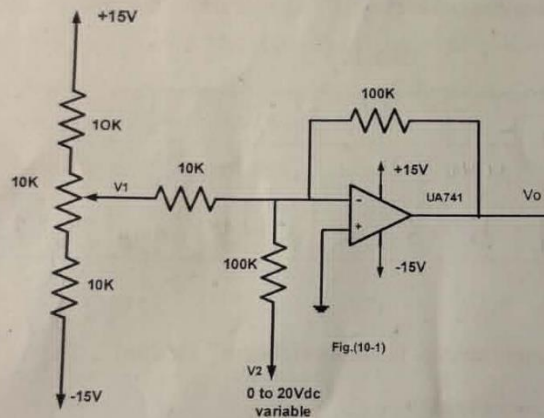
Prelab:

You have to apply PSPICE simulation to all practical circuits shown in the procedure below, and you have to do all the necessary calculations you will need in the lab.

PROCEDURE:

1. Adding Application

- Set up the circuit of Fig.(10-1), V1 is controlled by the potentiometer and V2, is obtained from the variable dc source on the trainer.



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- Measure the output voltage for V1, V2 as shown in table 10.1.

Table 10.1

Input voltage		Output voltage	
V ₁	V ₂	V _o	Calculated voltage
0.50498	21.988	-6.88	
0.30301	44.018	-6.922	
-0.9 -0.8983	22.006	6.858	
-1.3 -1.5192	66.008	8.995	

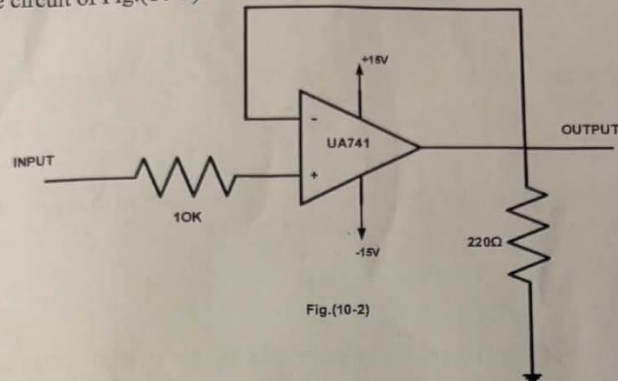
- Calculate the expected output voltage for each step using the formula :

$$V_o = XV_1 + YV_2$$

where X, Y is the resistors ratios.

II. Voltage Follower Application

4. Set up the circuit of Fig.(10-2).



5. Measure and records V_o for a dc input $V_i = (1V, 2V, 3V, 4V, 5V, 6V, 7V, 8, 10, 12, 14 \text{ and } 15)$.
 6. Change R_L (220Ω) to $1K\Omega$, then measure and record V_o for the same values above as shown in Table 10.2.

Table 10.2

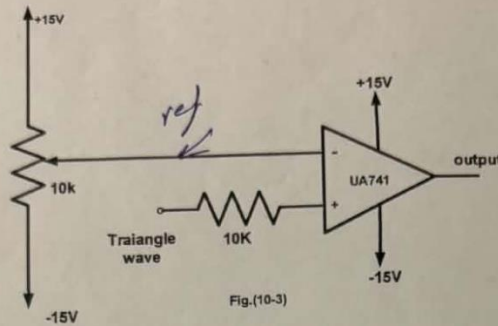
V_i [DC,V]	1	3	4	5	6	7	8	10	12	14	15
V_o For $R_L=220\Omega$	1	3	4	4.42	4.42						
V_o For $R_L=1k\Omega$	1	3	4	5	6	7	8	10	12	12.2	12.22

Question:

- Does this circuit have similar properties as the emitter follower? Explain?
- For what applications is this circuit used?
- What is the relation between your V_i , and V_o ?
- What is the approximate value of the maximum output current of the op-amp?

III. Comparator Application

1. Set up the circuit of Fig.(10-3).



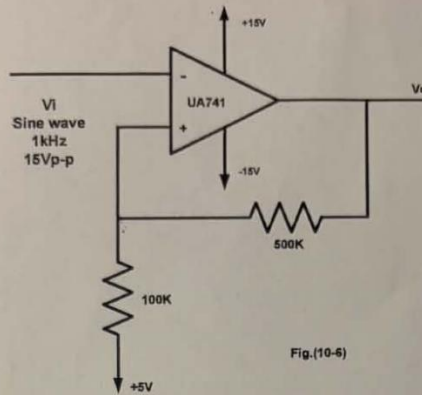
2. Use 1 kHz triangular input signal from the function generator.
3. Set the triangle input signal to 2 Vp-p and change the **dc reference voltage** in small steps from -1.5 to +1.5 so that you obtain an output of positive +Vsat then negative -Vsat and a square wave output.
4. For each of these cases draw the output voltage record the value of the dc reference voltage and take a picture of the scope screen.

Question:

- What happens to the amplifier output? For what application is this circuit used?
- Is there any similarity between this circuit and the diff amplifier, what is the shape of the output?

IV. Comparator with Hysteresis (Schmitt Trigger):

1. Connect the **Schmitt trigger** circuit shown in Fig.(10-6).



2. Put $V_i(t) = 15V_{p-p}$ sine wave of frequency 1 kHz.
3. Sketch the output voltage with respect to $V_i(t)$.
4. Indicate the levels of $V_i(t)$ where $V_o(t)$ changes its level.
5. Calculate the theoretical lower and upper trigger levels for the circuit above and compare them with those of measured values.
6. On the oscilloscope screen show the V_o as a function of V_i using XY format

